Parishram 2.0 2026 (Class 12th)

Biology (Botany)

DPP: 08

Sexual reproduction in flowering plants

Q1 What is the formation of seeds without fertilisation in certain flowering plants called?
(A) Parthenogenesis
(B) Polyembryony
(C) Apomixis
(D) Parthenocarpy

Q2 In apomixis, the diploid egg cell develops into an embryo without fertilisation. What is the genetic nature of the embryo?
(A) Genetically different from the parent plant
(B) Genetically identical to the parent plant
(C) A hybrid of two different plants
(D) A clone of the egg cell only

Q3 Which of the following is a key benefit of apomixis in the agricultural industry?
(A) Reduction in seed cost for farmers
(B) Increased genetic diversity
(C) Increased hybrid segregation
(D) Shortened growing season

Q4 Apomixis is found in some species of;
(A) Rosaceae and Citrus.
(B) Asteraceae and grasses.
(C) Leguminoseae and coconut.
(D) Solanaceae and Asteraceae.

Q5 What is the major advantage of incorporating apomixis into hybrid crop varieties?
(A) It enhances genetic diversity in the progeny.
(B) It eliminates the need for cross-pollination.
(C) It allows farmers to reuse hybrid seeds without loss of desired traits.
(D) It increases the rate of mutation in offspring.

Q6 **Assertion:** Apomixis can result in the formation of multiple embryos in a seed.
Reason: In species like Citrus and Mango, multiple embryos develop from cells surrounding the embryo sac.
(A) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion.
(B) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion.
(C) Assertion is correct but the Reason is incorrect.
(D) Both Assertion and Reason are incorrect.

Q7 Given below are two statements:
Statement I: The transformation of ovules into seeds and ovary into fruit proceeds simultaneously.
Statement II: Parthenocarpy can be induced through the application of growth hormones.
In the light of the above statements, choose the most appropriate answer from the options given below:
(A) Statement I is correct but Statement II is incorrect.
(B) Statement I is incorrect but Statement II is correct.
(C) Both Statement I and Statement II are correct.
(D) Both Statement I and Statement II are incorrect.

Q8 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Apple, strawberry and cashew are called true fruits.

Reason R: In apple, strawberry and cashew the thalamus also contributes to fruit formation.

In the light of the above statements, choose the **correct** answer from the options given below:

(A) A is true but R is false.

(B) A is false but R is true.

(C) Both A and R are true and R is the correct explanation of A.

(D) Both A and R are true but R is NOT the correct explanation of A.

Q9 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Hybrid varieties of several of our food and vegetable crops are being extensively cultivated.

Reason R: One of the benefits of hybrids is that hybrid seeds have to be produced every year.

In the light of the above statements, choose the **correct** answer from the options given below:

(A) A is true but R is false.

(B) A is false but R is true.

(C) Both A and R are true and R is the correct explanation of A.

(D) Both A and R are true but R is NOT the correct explanation of A.

Q10 Read the following statements and select the correct one(s):

A. Apomixis has been reported in some species of Asteraceae and grasses.

B. Apomixis is a form of sexual reproduction that mimics asexual reproduction.

C. In *Citrus* and mango, cells of the nucellus start dividing, protrude into embryo sac and develop into the embryos.

D. Occurrence of more than one embryo in a wall of fruit is referred to as polyembryony.

Choose the **correct** option:

(A) A and D only (B) B and D only

(C) Only A (D) A and C only

Answer Key

Q1	C
Q2	B
Q3	A
Q4	B
Q5	C
Q6	A
Q7	C
Q8	B
Q9	A
Q10	D

Hints & Solutions

Note: scan the QR code to watch video solution

Q1 **Text Solution:**

Apomixis

Video Solution:

Q2 **Text Solution:**

In apomixis, the diploid egg cell develops into an embryo without fertilisation. Since the egg cell is diploid (having the same number of chromosomes as the parent plant) and does not undergo meiosis, the resulting embryo is genetically identical to the parent plant. Therefore, the embryo produced through apomixis is genetically identical to the parent.

Video Solution:

Q3 **Text Solution:**

Apomixis allows the production of seeds that are genetically identical to the parent plant. This means farmers can save seeds from one generation and use them for the next without the loss of hybrid vigour or the need to purchase new seeds every year, thus reducing seed costs. The other options (B, C, D) are not key benefits of apomixis in agriculture.

Video Solution:

Q4 **Text Solution:**

Although seeds, in general are the products of fertilisation, a few flowering plants such as some species of Asteraceae and grasses, have evolved a special mechanism, to produce seeds without fertilisation, called apomixis.

Video Solution:

Q5 **Text Solution:**

It allows farmers to reuse hybrid seeds without loss of desired traits.

Video Solution:

Q6 **Text Solution:**

Both Assertion and Reason are correct and Reason is the correct explanation for Assertion.

Video Solution:

Q7 Text Solution:

- The transformation of ovules into seeds and ovary into fruit proceeds simultaneously.
- Parthenocarpy can be induced through the application of growth hormones.

Video Solution:

Q8 Text Solution:

Apple, strawberry and cashew are called false fruits.

Video Solution:

Q9 Text Solution:

Hybrid varieties of several of our food and vegetable crops are being extensively cultivated.

One of the problems of hybrids is that hybrid seeds have to be produced every year.

Video Solution:

Q10 Text Solution:

A. Apomixis has been reported in some species of Asteraceae and grasses.

- Apomixis occurs naturally in some species of Asteraceae and grasses.
- This is a well-known example of asexual reproduction mimicking seed formation.

B. Apomixis is a form of sexual reproduction that mimics asexual reproduction.

- This statement is incorrectly worded.
- Correct understanding: Apomixis is a form of asexual reproduction that mimics sexual reproduction (i.e., seed formation without fertilization).
- Hence, this statement is reversed.

C. In *Citrus* and mango, cells of the nucellus may develop into embryo sacs.

- In Citrus and mango, nucellar embryony is a type of apomixis where nucellus cells (diploid) form embryos without fertilization.
- These develop additional embryos alongside the zygotic embryo.

D. Occurrence of more than one embryo in a seed is referred to as polyembryony.

- Polyembryony is defined as the presence of more than one embryo in a seed.

Video Solution:





[Android App](#) | [iOS App](#) | [PW Website](#)